$$-\frac{1}{4}p_0SL\left(1+\frac{\rho g(L-h)}{p_0}\right)\sqrt{\left(1+\frac{h}{L}+\frac{p_0}{\rho gL}\right)^2-\frac{4h}{L}}. \quad (16)$$

$$Q = 17.0\ kJ. \quad (17)$$

**채점 기준**

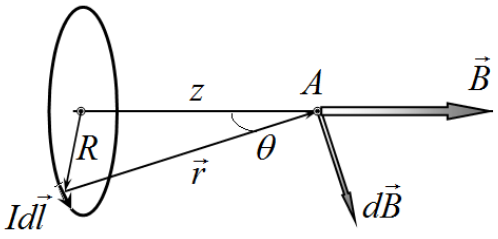
| № | 평가 항목 | 배점 | |
|---|---|---|---|
| 1 | (1) $p = p_0 + \rho g(L - x_0)$ | 0,25 | **2,0** |
| | (2) $p_0Sh = vRT_0$ | 0,25 | |
| | (3) $pS(h - x_0) = vRT_0$ | 0,25 | |
| | (4) $\rho g x_0^2 - [p_0 + \rho g(L + h)]x_0 + \rho ghL = 0$ | 0,25 | |
| | (5) $x_0 = \frac{1}{2}\left[\frac{p_0}{\rho g} + L + h \pm \sqrt{\left(\frac{p_0}{\rho g} + L + h\right)^2 - 4hL}\right]$ | 0,25 | |
| | (6) $x_0 = \frac{1}{2}\left[\frac{p_0}{\rho g} + L + h - \sqrt{\left(\frac{p_0}{\rho g} + L + h\right)^2 - 4hL}\right]$ | 0,5 | |
| | (7) $x_0 = 7{,}86 \cdot 10^{-2}\ m$ | 0,25 | |
| 2 | (8) $p(x) = p_0 + \rho g(L - x)$ | 0,5 | **0,5** |
| 3 | (9) $p(x)S(h - x) = vRT(x)$ | 0,5 | **1,0** |
| | (10) $T(x) = T_0\left(1 - \frac{x}{h}\right)\left(1 + \frac{\rho g(L-x)}{p_0}\right)$ | 0,5 | |
| 4 | (11) $T_m = T_0\left(1 + \frac{\rho gL}{p_0}\right)$ | 0,5 | **1,0** |
| | (12) $T_m = 350\ K$ | 0,5 | |
| 5 | (13) $\Delta U = \frac{5}{2}vR(T - T_0) = \frac{5}{2}\rho gLSh$ | 0,5 | **2,5** |
| | (14) $A = \frac{1}{2}p_0SL\left(1 + \frac{p_0}{2\rho gL} + \frac{\rho gL}{2p_0}\left[1 + \frac{2h}{L} - \frac{h^2}{L^2}\right]\right) - \frac{1}{4}p_0SL\left(1 + \frac{\rho g(L-h)}{p_0}\right)\sqrt{\left(1 + \frac{h}{L} + \frac{p_0}{\rho gL}\right)^2 - \frac{4h}{L}}.$ | 0,5 | |
| | (15) $Q = \Delta U + A$ | 0,5 | |
| | (16) $Q = \frac{1}{2}p_0SL\left(1 + \frac{p_0}{2\rho gL} + \frac{\rho gL}{2p_0}\left[1 + \frac{12h}{L} - \frac{h^2}{L^2}\right]\right) - \frac{1}{4}p_0SL\left(1 + \frac{\rho g(L-h)}{p_0}\right)\sqrt{\left(1 + \frac{h}{L} + \frac{p_0}{\rho gL}\right)^2 - \frac{4h}{L}}.$ | 0,5 | |
| | (17) $Q = 17{,}0\ kJ$ | 0,5 | |
| **합계** | | | **7,0** |

## 문제 3. 지연과 감쇠 (16점)

1.1.1

.

. $z$




$$dB_z = \frac{\mu_0}{4\pi}\frac{Idl}{r^2}\sin\theta \quad (1)$$

$$dB_z = \frac{\mu_0}{4\pi}\frac{Idl}{r^2}\frac{R}{r} = \frac{\mu_0}{4\pi}\frac{IdlR}{r^3} = \frac{\mu_0}{4\pi}\frac{IdlR}{\left(R^2+z^2\right)^{\frac{3}{2}}} \quad (2)$$